Collision energy revisited

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Learning target: Use energy as well as momentum to classify a collision.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 14; 3 course days later.*
15 N toward the natural length.

02. Explain a worked example (worked)
Total momentum 2 both times; K 2 to 1 J, remainder internal/thermal/sound.

03. Look and explain (model)
Total momentum zero both before after; kinetic energy decreases to zero after sticking.

04. Build the solution (practice)
Each 2 J; total 4 J; final 0 J.

05. Use the idea (reason)
Sticking is only perfectly inelastic; bouncing can still transform kinetic energy.

06. Choose a method (mixed) *Review: Unit 4, Day 5; 12 course days later.*
32 J, using $K_f=K_i+W_{net}$.

07. Exit check (exit)
Inelastic; thermal, deformation, sound, etc.

08. STRETCH: Apply it another way (stretch)
Final zero; initial 4 J. Converted into internal energy and other transfers, not destroyed.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 14; 3 course days later.*
F=-kx=+7.5 N, right.

02. Explain a worked example (worked)
Total momentum 2 both times; K 2 to 1 J, remainder internal/thermal/sound.

03. Represent and interpret (model)
Total momentum zero both before after; kinetic energy decreases to zero after sticking.

04. Solve and justify (practice)
vf 2 m/s; 9 J; 6 J; 3 J transformed.

05. Analyze the reasoning (reason)
Sticking is only perfectly inelastic; bouncing can still transform kinetic energy.

06. Choose a method (mixed) *Review: Unit 4, Day 5; 12 course days later.*
Net work 72 J; v_f=6 m/s from ½mv²=W_{net}.

07. Exit check (exit)
Momentum 4 both; K before 8 J, after 5 J; inelastic.

08. STRETCH: Apply it another way (stretch)
v_f=2 m/s. K_i=18 J; K_f=6 J; loss 12 J or 2/3.